
Discrete Channel Capacity

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Channel capacity



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A general formula for channel capacity

[PDF] researchgate.net

[S Verdú](#) - IEEE Transactions on Information Theory, 1994 - [ieeexplore.ieee.org](#)

... and channel capacity are based on ... channel capacity? At first glance, it may seem that the answer is negative because the channel encoder (unlike the source encoder or the channel de...

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Channel coding: The road to channel capacity

[PDF] arxiv.org

[DJ Costello](#), [GD Forney](#) - Proceedings of the IEEE, 2007 - [ieeexplore.ieee.org](#)

... channel coding theorem, we trace the evolution of channel coding from Hamming codes to capacity... on the additive white Gaussian noise channel. We discuss algebraic block codes, ...

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Estimate of channel capacity in Rayleigh fading environment

[WCY Lee](#) - IEEE transactions on Vehicular Technology, 1990 - [ieeexplore.ieee.org](#)

... channel capacity in a Rayleigh fading environment has been derived. The result shows that the channel capacity ... what the degradations are in channel capacity due to Rayleigh fading, ...

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Interactive channel capacity

[PDF] weizmann.ac.il

[G Kol](#), [R Raz](#) - Proceedings of the forty-fifth annual ACM symposium ..., 2013 - [dl.acm.org](#)

... In this paper we study the interactive channel capacity of the binary symmetric channel. For ... the binary symmetric channel with noise rate ϵ . We define the channel capacity $C(\epsilon)$ by1 ...

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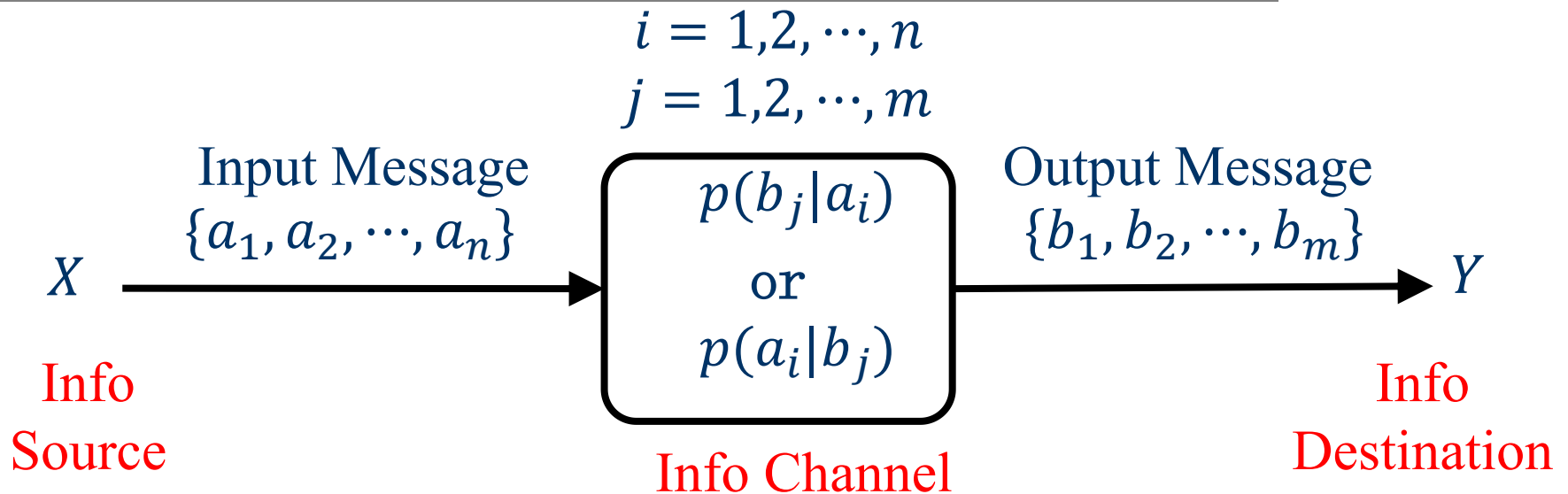
Main Content

1 Definition of Channel Capacity

2 Capacity of Classic Channels

3 Capacity of Multi-Symbol Channel

Single Symbol Discrete Channel



- ❑ The statistical property of the info channel can be characterised by the conditional probabilities $\{p(b_j|a_i)|i = 1, 2, \dots, n, j = 1, 2, \dots, m\}$, which are also regarded as the channel transition probabilities.
- ❑ It can also be characterized by the conditional probabilities $\{p(a_i|b_j)|i = 1, 2, \dots, n, j = 1, 2, \dots, m\}$, which are also regarded as the reverse channel transition probabilities

Channel Transition Matrix

- Let rows represent the channel input (info source) X and columns represent the channel output (info dest) Y , the **channel transition matrix** is

$$\begin{bmatrix} p(b_1/a_1) & p(b_2/a_1) & \cdots & p(b_n/a_1) \\ p(b_1/a_2) & p(b_2/a_2) & \cdots & p(b_n/a_2) \\ \vdots & \vdots & \ddots & \vdots \\ p(b_1/a_n) & p(b_2/a_n) & \cdots & p(b_n/a_n) \end{bmatrix},$$

which characterises the statistical property of Y by knowing X .

- Let rows represent the channel output (info dest) Y and columns represent the channel input (info source) X , the **reverse channel transition matrix** is

$$\begin{bmatrix} p(a_1/b_1) & p(a_2/b_1) & \cdots & p(a_n/b_1) \\ p(a_1/b_2) & p(a_2/b_2) & \cdots & p(a_n/b_2) \\ \vdots & \vdots & \ddots & \vdots \\ p(a_1/b_n) & p(a_2/b_n) & \cdots & p(a_n/b_n) \end{bmatrix}$$

which characterises the statistical property of X by knowing Y .

Definition of Channel Capacity

- Mutual information between the info source X and the info dest Y is the amount of info transmitted in the channel (or info transmission rate)

$$R = I(X; Y) \text{ bit/symbol}$$

- When the channel is given, $I(X; Y)$ is a function of the source's probability distribution $\{p(a_i)\}$. Since $I(X; Y)$ is a concave function of $\{p(a_i)\}$, we may find the optimal $\{p^*(a_i)\}$ for maximising the mutual info $I(X; Y)$. This maximum mutual info is defined as the channel capacity C :

$$C = \max_{\{p(a_i)\}} R = \max_{\{p(a_i)\}} I(X; Y) \text{ bit/symbol}$$

If the transmission time of a symbol is t seconds, the channel capacity per second is

$$C_t = \frac{1}{t} \max_{\{p(a_i)\}} I(X; Y) \text{ bit/s}$$

Channel capacity is only determined by the info channel itself:

$$C = \max_{\{p(a_i)\}} I(X; Y) = I(\{p^*(a_i)\}, \{p(b_j/a_i)\}) \text{ bit/sign}$$

Mutual Info and Channel Capacity

□ Observe from info dest, the mutual info of the mutli-symbol channel

is

$$I(\mathbf{X}; \mathbf{Y}) = \sum_{i=1}^{n^N} \sum_{j=1}^{m^M} p(\alpha_i \beta_j) \log_2 \frac{p(\alpha_i / \beta_j)}{p(\alpha_i)}$$
$$= \sum_{i_1=1}^n \cdots \sum_{i_N=1}^n \sum_{j_1=1}^m \cdots \sum_{j_M=1}^m p(a_{i_1} \cdots a_{i_N} b_{j_1} \cdots b_{j_M}) \log_2 \frac{p(a_{i_1} \cdots a_{i_N} / b_{j_1} \cdots b_{j_M})}{p(a_{i_1} \cdots a_{i_N})}$$

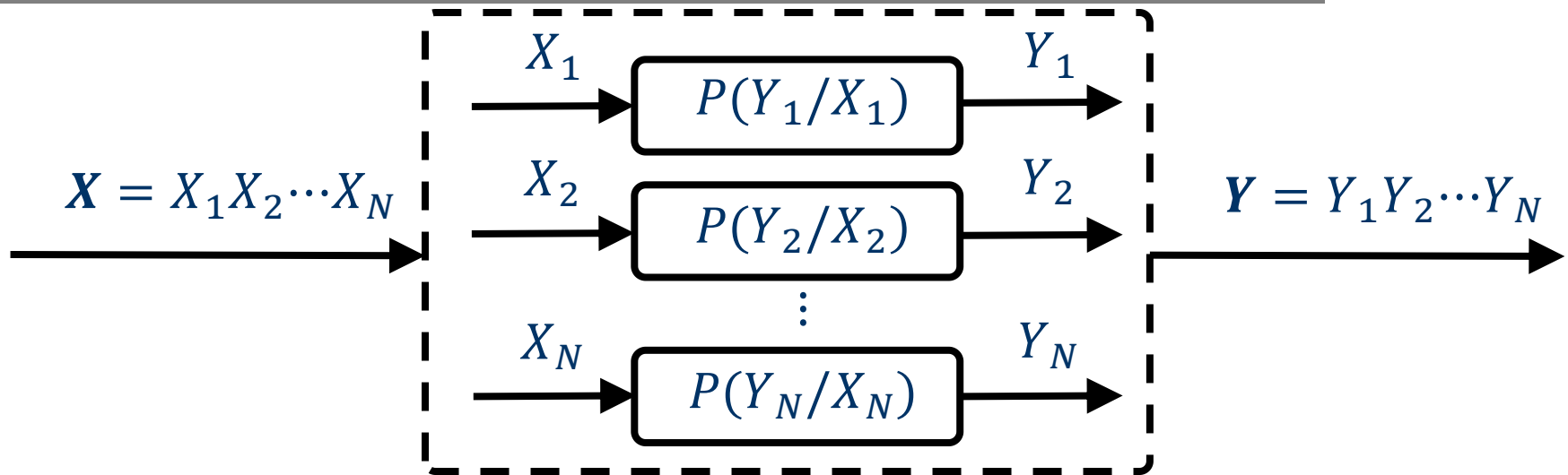
□ Observe from info source, its mutual info is

$$I(\mathbf{X}; \mathbf{Y}) = \sum_{i=1}^{n^N} \sum_{j=1}^{m^M} p(\alpha_i \beta_j) \log_2 \frac{p(\beta_j / \alpha_i)}{p(\beta_j)}$$

□ The capacity of the multi-symbol channel is

$$C = \max_{\{p(\alpha_i)\}} I(\mathbf{X}; \mathbf{Y}), \quad i = 1, \dots, n^N$$

Memoryless Expanded Channel



- ❑ Expanded channel with order N : a single-symbol channel independently operates N times at N different moments.
- ❑ Transition probabilities of the memoryless expanded channel:

$$p(\beta_j/\alpha_i) = p(b_{j_1} \cdots b_{j_N}/a_{i_1} \cdots a_{i_N})$$

$$= p(b_{j_1}/a_{i_1})p(b_{j_2}/a_{i_2}) \cdots p(b_{j_N}/a_{i_N}) = \prod_{k=1}^N p(b_{j_k}/a_{i_k})$$

Mutual Info of Expanded Channel

- Noise entropy of the **memoryless expanded** channel with order N is:

$$\begin{aligned} H(\mathbf{Y}|\mathbf{X}) &= H(Y_1 \cdots Y_N | X_1 \cdots X_N) \\ &= H(Y_1 | X_1) + H(Y_2 | X_2) + \cdots + H(Y_N | X_N). \end{aligned}$$

- Mutual info of the **memoryless expanded** channel with order N is:

$$I(\mathbf{X}; \mathbf{Y}) = H(\mathbf{Y}) - H(\mathbf{Y}|\mathbf{X}) = H(Y_1 \cdots Y_N) - \sum_{k=1}^N H(Y_k | X_k).$$

We have $I(\mathbf{X}; \mathbf{Y}) \leq \sum_{k=1}^N I(X_k; Y_k)$, where $I(X_k; Y_k)$ is the mutual info of the k -th sub channel. If the source is memoryless, the equality holds.

- The mutual info of the memoryless expanded channel with order N is $I(\mathbf{X}; \mathbf{Y}) = NI(X; Y)$ and the channel capacity is $C^N = NC$, if the source is memoryless expanded with order N .

Example

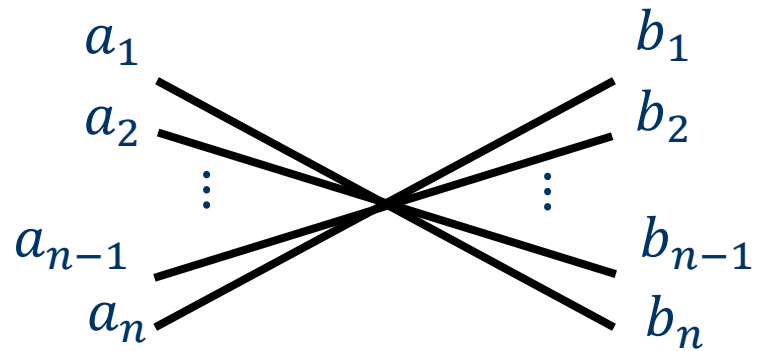
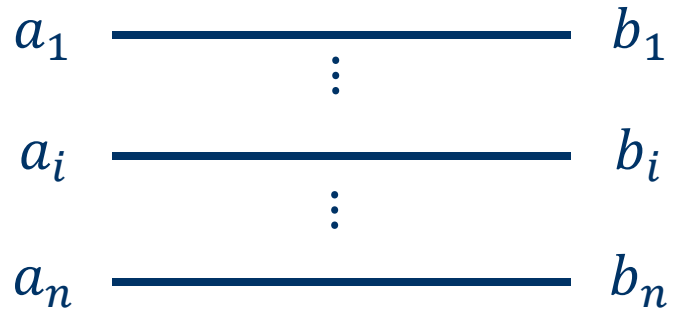
□ The transition matrix of a binary info channel is

$$p(Y/X) = \begin{bmatrix} 1 & 0 \\ 0.5 & 0.5 \end{bmatrix}$$

- Please calculate the mutual info $I(X; Y)$, when the source is equally distributed.
- Please calculate the channel capacity C and the corresponding info source probability distribution $p(X)$.

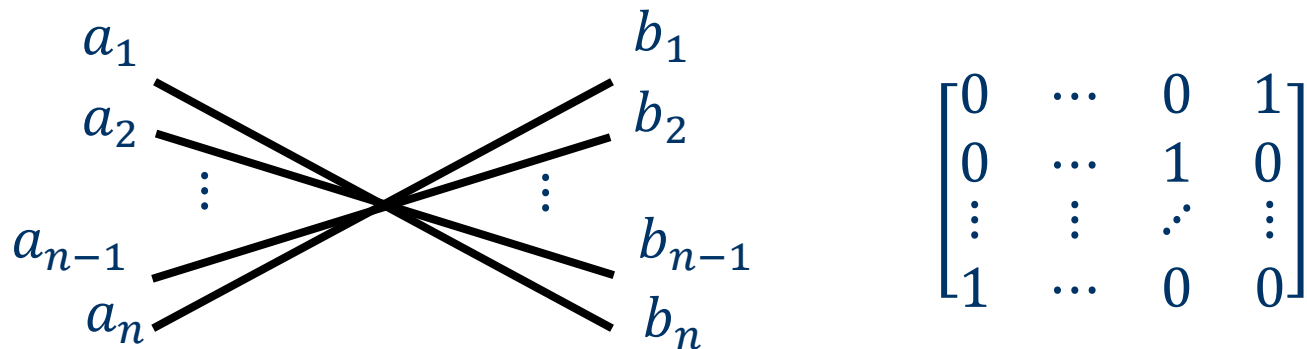
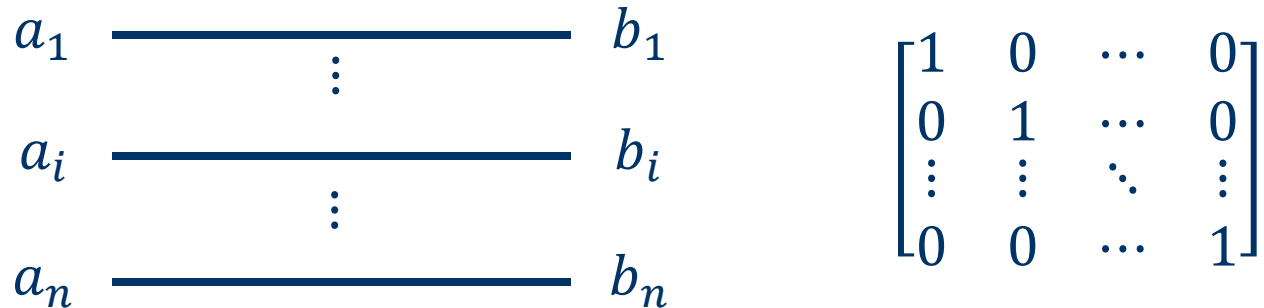
Solution: Exploit the definition of the channel capacity and maximise the mutual information by using its derivative with respect to the source's probability distribution.

Example 1: Noise? Loss?



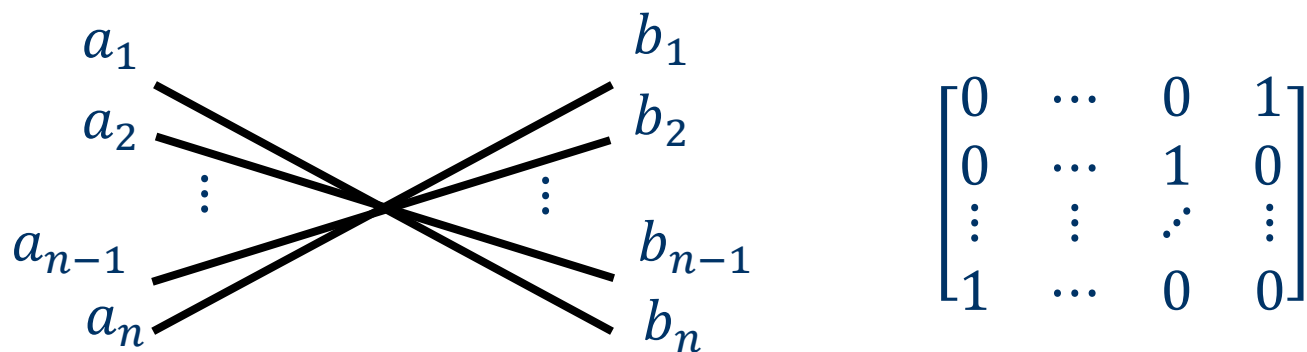
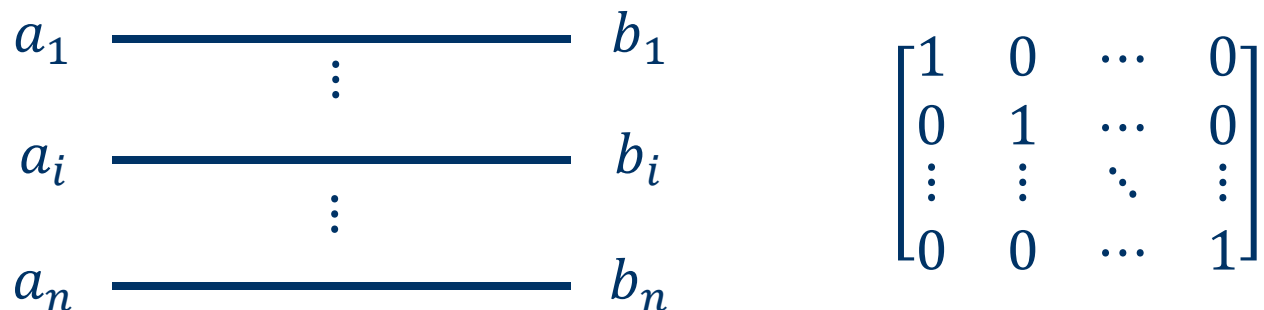
Discrete Noiseless Channel

□ Noiseless channel having one-on-one mapping between its input and output:



Discrete Noiseless Channel

- Noiseless channel having one-on-one mapping between its input and output:



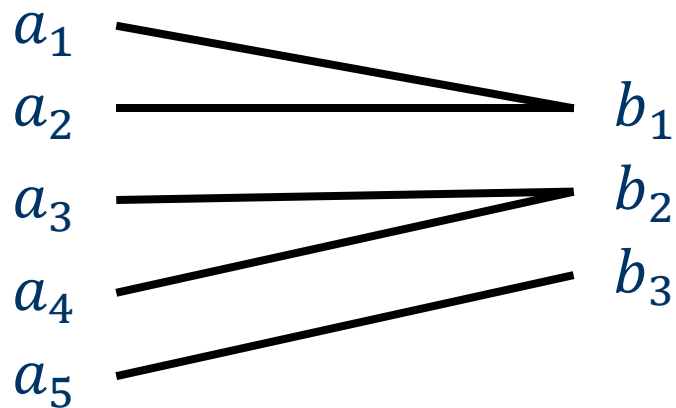
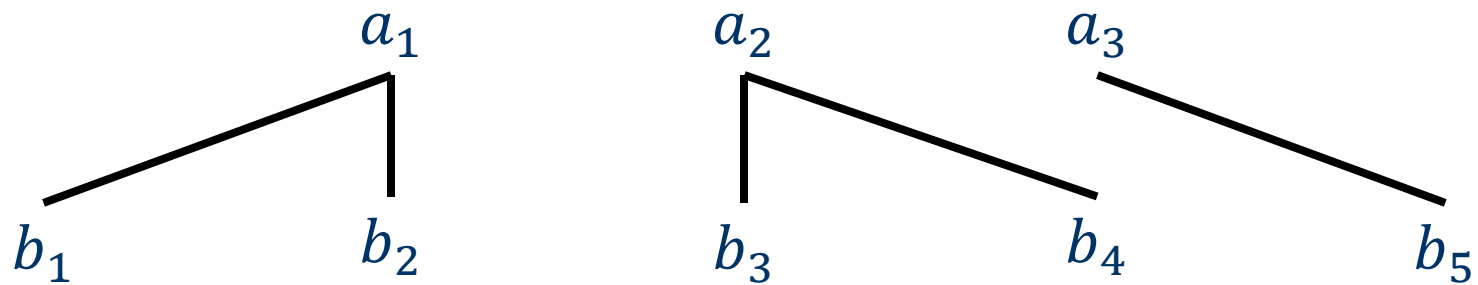
- When the source is equally distributed, the channel capacity is achieved, which is $C = \max_{\{p(a_i)\}} I(X; Y) = \max_{\{p(a_i)\}} H(X) = \log_2 n$

Discrete Noiseless Channel

- We can also calculate the information capacity $C = \max I(X; Y) = 1$ bit, which is achieved by using $p(x) = (1/2, 1/2)$.

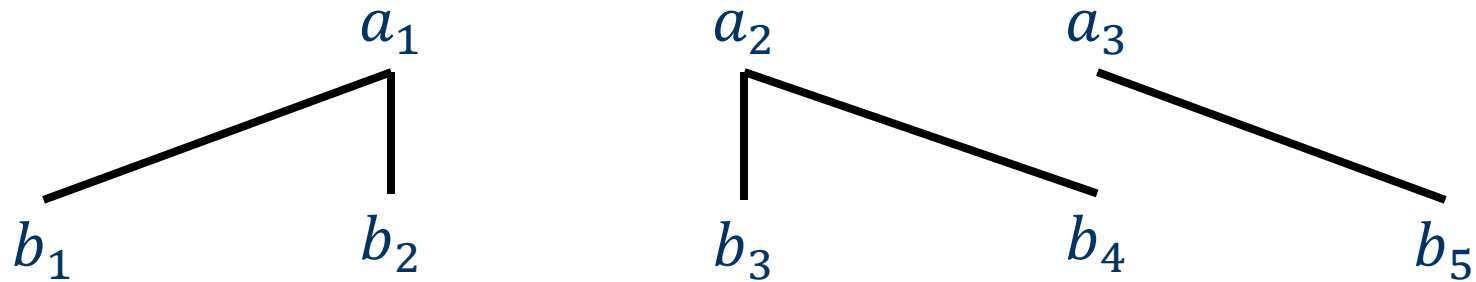


Example 2: Noise? Loss?



Discrete Noiseless Channel

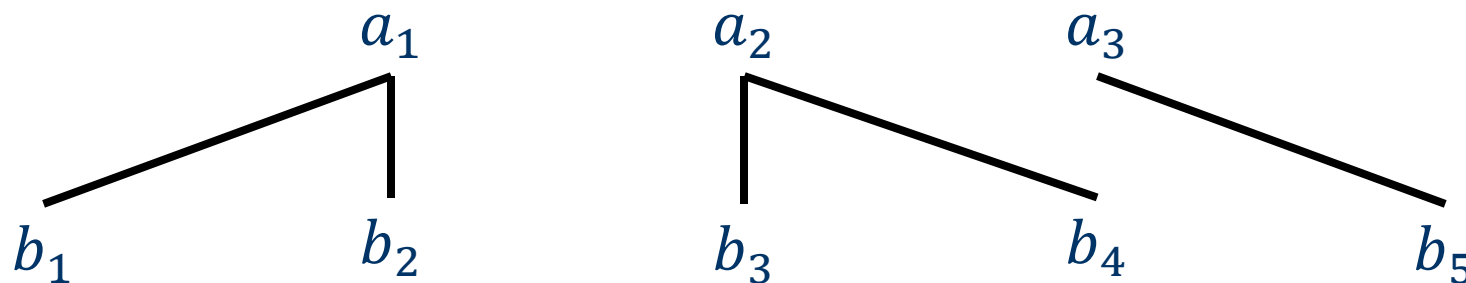
□ Noiseless channel having expansive function (non-overlapping):



$$\begin{bmatrix} p(b_1/a_1) & p(b_2/a_1) & 0 & 0 & 0 \\ 0 & 0 & p(b_3/a_2) & p(b_4/a_2) & 0 \\ 0 & 0 & 0 & 0 & p(b_5/a_3) \end{bmatrix}$$

Discrete Noiseless Channel

□ Noiseless channel having expansive function (non-overlapping):

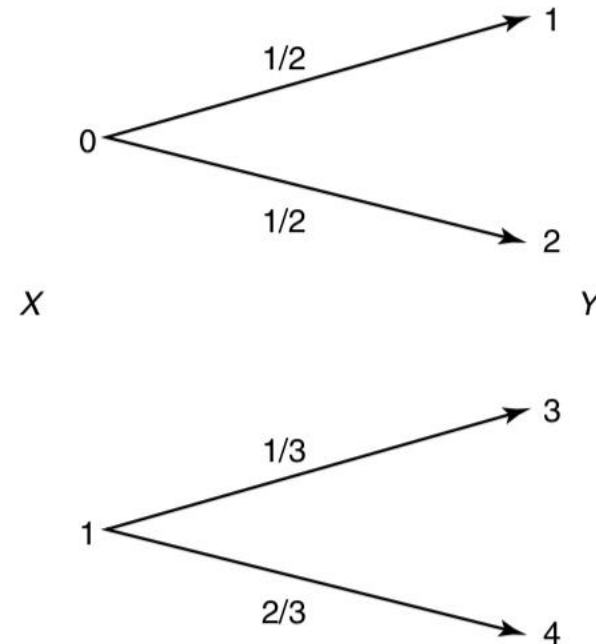


$$\begin{bmatrix} p(b_1/a_1) & p(b_2/a_1) & 0 & 0 & 0 \\ 0 & 0 & p(b_3/a_2) & p(b_4/a_2) & 0 \\ 0 & 0 & 0 & 0 & p(b_5/a_3) \end{bmatrix}$$

□ Channel capacity is $C = \max_{\{p(a_i)\}} I(X; Y) = \max_{\{p(a_i)\}} H(X) = \log_2 n$

Noisy Channel with Nonoverlapping Outputs

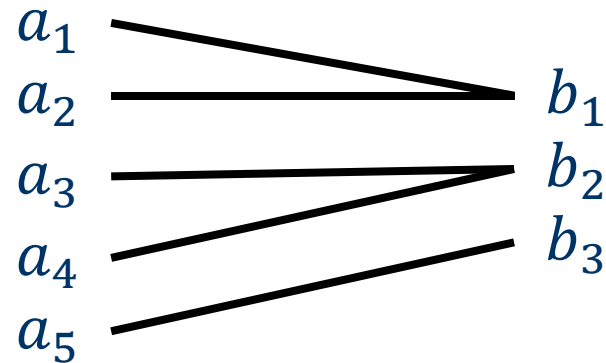
- We can also calculate the information capacity $C = \max I(X;Y) = 1$ bit, which is achieved by using $p(x) = (1/2, 1/2)$.



Discrete Noiseless Channel

□ Noiseless Channel having merging function:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

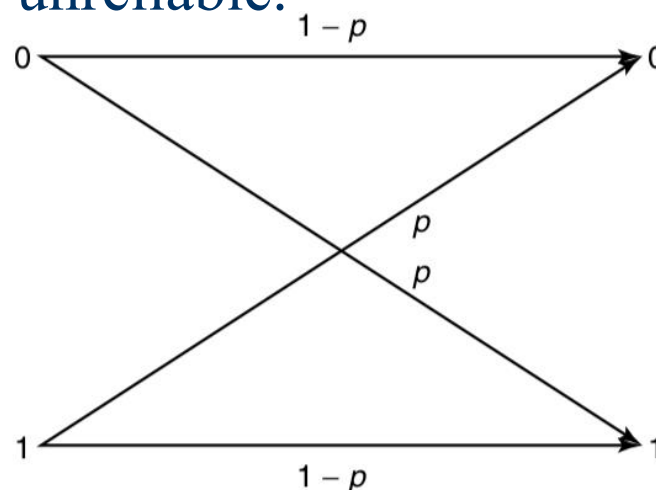


□ Channel capacity is $C = \max_{\{p(a_i)\}} I(X; Y) = \max_{\{p(a_i)\}} H(Y) = \log_2 m$

Binary Symmetric Channel

- This is the simplest model of a channel with errors, yet it captures most of the complexity of the general problem.
- In a sense, all the bits received are unreliable.

$$\begin{aligned} I(X; Y) &= H(Y) - H(Y|X) \\ &= H(Y) - \sum p(x)H(Y|X = x) \\ &= H(Y) - \sum p(x)H(p) \\ &= H(Y) - H(p) \\ &\leq 1 - H(p), \end{aligned}$$



where the last inequality follows because Y is a binary random variable. Equality is achieved when the input distribution is uniform.

- Hence, the information capacity of a binary symmetric channel with parameter p is $C = 1 - H(p)$ bits.

Symmetric Channel

- If both the rows and columns of a channel transition matrix are permutable, it is a symmetric channel.

Symmetric Channel $P_1 = \begin{bmatrix} 1/3 & 1/3 & 1/6 & 1/6 \\ 1/6 & 1/6 & 1/3 & 1/3 \end{bmatrix}$ $P_2 = \begin{bmatrix} 1/2 & 1/3 & 1/6 \\ 1/6 & 1/2 & 1/3 \\ 1/3 & 1/6 & 1/2 \end{bmatrix}$

Asymmetric Channel $P_3 = \begin{bmatrix} 1/3 & 1/3 & 1/6 & 1/6 \\ 1/6 & 1/3 & 1/6 & 1/3 \end{bmatrix}$ $P_4 = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.1 & 0.7 \end{bmatrix}$

- Channel Capacity is $C = \max_{\{p(a_i)\}} \{H(Y) - H(q_1, q_2, \dots, q_m)\}$
 $= \log_2 m - H(q_1, q_2, \dots, q_m),$

where $\{q_1, q_2, \dots, q_m\}$ are the transition probabilities in a row of the channel's transition matrix.

Strongly Symmetric Channel

- Now consider the channel with transition matrix:

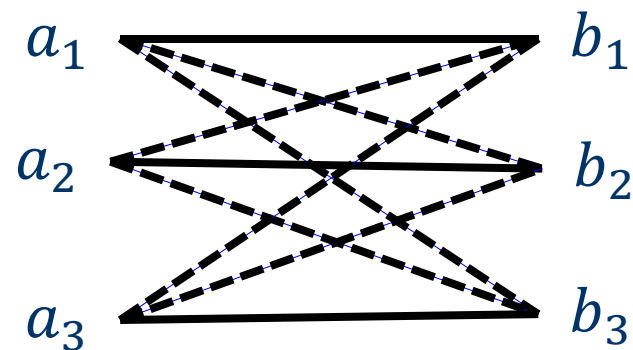
$$p(y|x) = \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{bmatrix}$$

- Here the entry in the x th row and the y th column denotes the conditional probability $p(y|x)$ that y is received when x is sent.
- all the rows of the probability transition matrix are permutations of each other and so are the columns.
- Such a channel is said to be *symmetric*.
- Other example: $Y=X+Z \pmod{c}$, where Z has some distribution on the integers $\{0, 1, 2, \dots, c-1\}$, X has the same alphabet as Z , and Z is independent of X .

Strongly Symmetric Channel

- In a strongly symmetric channel, every transmit symbol has a probability of $\bar{p}$ to be correctly received and it has a probability of $\frac{p}{n-1}$ to be wrongly received as any other symbol:

$$P_{n \times n} = \begin{bmatrix} \bar{p} & \frac{p}{n-1} & \cdots & \frac{p}{n-1} \\ \frac{p}{n-1} & \bar{p} & \cdots & \frac{p}{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{p}{n-1} & \frac{p}{n-1} & \cdots & \bar{p} \end{bmatrix}$$



To find an explicit expression for the capacity of the channel, letting $\mathbf{r}$ be a row of the transition matrix, we have

$$I(X; Y) = H(Y) - H(Y|X)$$

$$= H(Y) - H(\mathbf{r})$$

$$\leq \log |\mathcal{Y}| - H(\mathbf{r})$$

with equality if the output distribution is uniform. But $p(x) = 1/|\mathcal{X}|$ achieves a uniform distribution on $\mathcal{Y}$.

Strongly Symmetric Channel(cont.)

$$p(y) = \sum_{x \in \mathcal{X}} p(y|x)p(x) = \frac{1}{|\mathcal{X}|} \sum p(y|x) = c \frac{1}{|\mathcal{X}|} = \frac{1}{|\mathcal{Y}|},$$

- where c is the sum of the entries in one column of the probability transition matrix.
- Thus for the channel with transition matrix below

$$p(y|x) = \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{bmatrix}$$

$$C = \max_{p(x)} I(X; Y) = \log 3 - H(0.5, 0.3, 0.2),$$

- C is achieved by a uniform distribution on the input.

Strongly Symmetric Channel(cont.)

$$P_{n \times n} = \begin{bmatrix} \bar{p} & \frac{p}{n-1} & \cdots & \frac{p}{n-1} \\ \frac{p}{n-1} & \bar{p} & \cdots & \frac{p}{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{p}{n-1} & \frac{p}{n-1} & \cdots & \bar{p} \end{bmatrix}$$

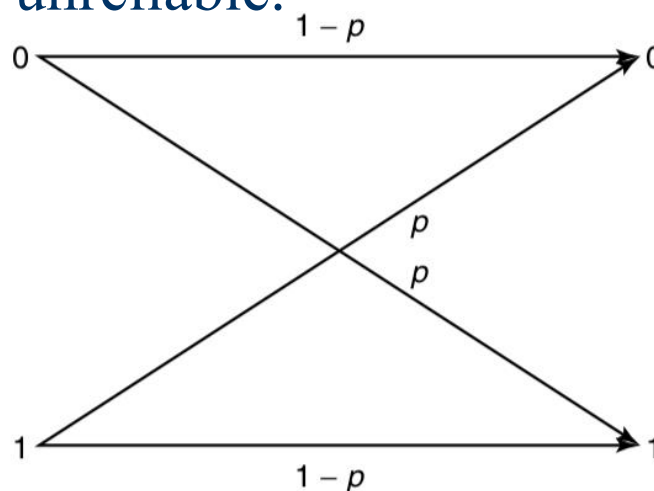
□ Channel Capacity is $C = \max_{\{p(a_i)\}} \left\{ H(Y) - H\left(\bar{p}, \frac{p}{n-1}, \dots, \frac{p}{n-1}\right) \right\}$

$$= \log_2 n + \bar{p} \log_2 \bar{p} + p \log_2 \frac{p}{n-1}$$

Review: Binary Symmetric Channel

- This is the simplest model of a channel with errors, yet it captures most of the complexity of the general problem.
- In a sense, all the bits received are unreliable.

$$\begin{aligned} I(X; Y) &= H(Y) - H(Y|X) \\ &= H(Y) - \sum p(x)H(Y|X = x) \\ &= H(Y) - \sum p(x)H(p) \\ &= H(Y) - H(p) \\ &\leq 1 - H(p), \end{aligned}$$



where the last inequality follows because Y is a binary random variable. Equality is achieved when the input distribution is uniform.

- Hence, the information capacity of a binary symmetric channel with parameter p is $C = 1 - H(p)$ bits.

Example

- The channel input is the 26 English letters $\{A, B, \dots, Z\}$. Every letter has a probability of $\frac{1}{2}$ to be successfully output by the channel. Every letter has a probability of $\frac{1}{2}$ to be erroneously output as the next letter (e. g. $A \rightarrow B$, $B \rightarrow C$, ... $Z \rightarrow A$). Please calculate the channel capacity and the corresponding optimal probability distribution of the information source.

Example

□ The transition matrix of a binary info channel is

$$p(Y/X) = \begin{bmatrix} 1 & 0 \\ 0.5 & 0.5 \end{bmatrix}$$

- Please calculate the mutual info $I(X; Y)$, when the source is equally distributed.
- Please calculate the channel capacity C and the corresponding info source probability distribution $p(X)$.

Solution: Exploit the definition of the channel capacity and maximise the mutual information by using its derivative with respect to the source's probability distribution.

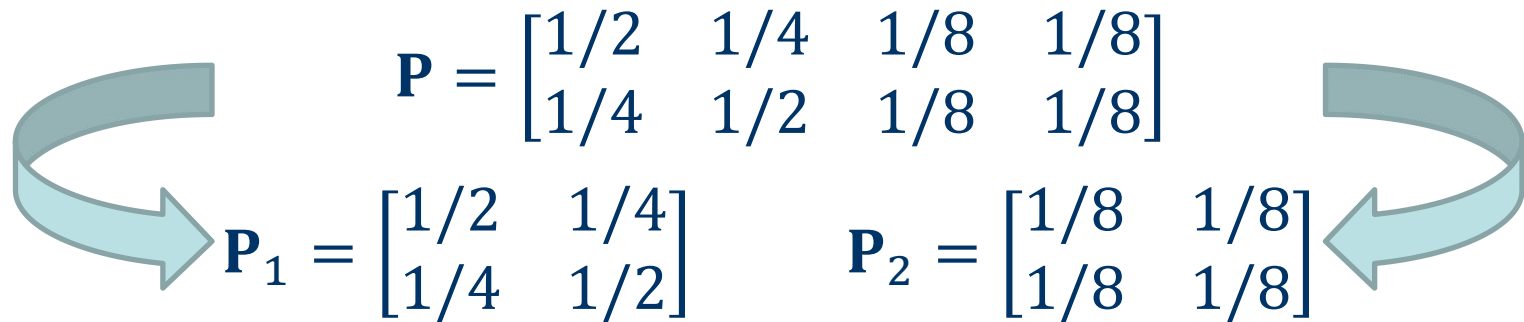
Weakly Symmetric Channel

- ❑ For a $n \times m$ channel transition matrix $\mathbf{P}$, its rows are permutable but its columns are not. And all the column sum $\sum_x p(y|x)$ are equal.
- ❑ Example: The channel with transition matrix below is weakly symmetric but not symmetric.

$$p(y|x) = \begin{pmatrix} \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}$$

Weakly Symmetric Channel

- However, its columns can be divided into s different subsets without any intersection. These subsets correspond to $m_1, m_2, \dots, m_s$ different channel output symbols.
- If the $n \times m_k$ sub-matrix $\mathbf{P}_k$'s rows and columns are both permutable, the channel correspond to its sub-matrix is a weakly symmetric channel.


$$\mathbf{P} = \begin{bmatrix} 1/2 & 1/4 & 1/8 & 1/8 \\ 1/4 & 1/2 & 1/8 & 1/8 \end{bmatrix}$$
$$\mathbf{P}_1 = \begin{bmatrix} 1/2 & 1/4 \\ 1/4 & 1/2 \end{bmatrix} \quad \mathbf{P}_2 = \begin{bmatrix} 1/8 & 1/8 \\ 1/8 & 1/8 \end{bmatrix}$$

□ When the source is equally distributed, the channel capacity is

$$C = - \sum_{k=1}^s m_k \bar{p}(M_k) \log_2 \bar{p}(M_k) - H(q_1, q_2, \dots, q_m)$$

Weakly Symmetric Channel

□ In the transition matrix of a weakly symmetric channel, its rows are the permutations of the transition probabilities $\{q_1, q_2, \dots, q_m\}$.

Step 1:

Divide the columns of the channel transition matrix into s sub-matrix, whose rows and columns are permutable. A sub-matrix corresponds to m_k , $k = 1, 2, \dots, s$ channel output symbols.

Step 2:

For every sub-matrix, calculate the probabilities of the corresp. channel output symbols. Calculate the average $\bar{p}(M_k)$ of these probabilities, when the source is equally distributed.

Step 3:

$$C = - \sum_{k=1}^s m_k \bar{p}(M_k) \log_2 \bar{p}(M_k) - H(q_1, q_2, \dots, q_m)$$

Example

- The probability distribution of a binary info source is $p(X) = \{p(0) = 0.8, p(1) = 0.2\}$. The transition probabilities of the channel is $p(Y/X) = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.1 & 0.3 & 0.6 \end{bmatrix}$.
- Please calculate the mutual info $I(X; Y)$.
 - Please calculate the channel capacity if the probability distribution of a binary info source is adjustable, and find out the corresponding optimal probability distribution of the binary info source $p^*(X)$.

$$C = - \sum_{k=1}^s m_k \bar{p}(M_k) \log_2 \bar{p}(M_k) - H(q_1, q_2, \dots, q_m)$$

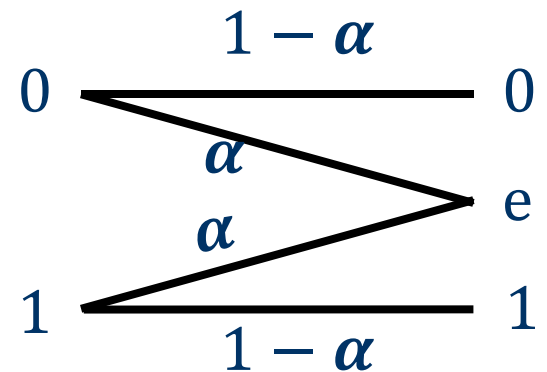
Example: Binary Erasure Channel

- In a binary erasure channel, the receiver either correctly receives the transmit symbol or discards the contaminated symbol.

Channel Transition Matrix

$$\begin{array}{c} 0 \quad e \quad 1 \\ 1 \end{array} \begin{bmatrix} 1 - \alpha & \alpha & 0 \\ 0 & \alpha & 1 - \alpha \end{bmatrix}$$

Channel Transition Graph



$$H(Y) = H((1 - \pi)(1 - \alpha), \alpha, \pi(1 - \alpha)) = H(\alpha) + (1 - \alpha)H(\pi).$$

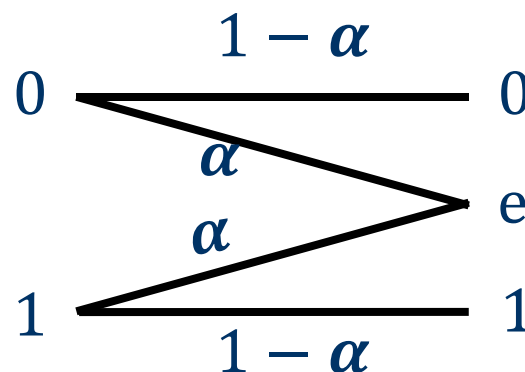
Example: Binary Erasure Channel

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Channel Transition Matrix

$$\begin{array}{c} 0 \quad e \quad 1 \\ \begin{array}{c} 0 \\ 1 \end{array} \quad \begin{bmatrix} 1-\alpha & \alpha & 0 \\ 0 & \alpha & 1-\alpha \end{bmatrix} \end{array}$$

Channel Transition Graph



$$C = \max_{p(x)} I(X; Y)$$

$$= \max_{p(x)} (H(Y) - H(Y|X))$$

$$= \max_{p(x)} H(Y) - H(\alpha).$$

Letting E be the event $\{Y = e\}$, using the

$$H(Y) = H(Y, E) = H(E) + H(Y|E)$$

and letting $\Pr(X = 1) = \pi$, we have

Example: Binary Erasure Channel (cont)

- Hence, the information capacity for the BEC

$$\begin{aligned} C &= \max_{p(x)} H(Y) - H(\alpha) \\ &= \max_{\pi} (1 - \alpha)H(\pi) + H(\alpha) - H(\alpha) \\ &= \max_{\pi} (1 - \alpha)H(\pi) \\ &= 1 - \alpha, \end{aligned}$$

where capacity is achieved by $\pi = 1/2$

- The expression for the capacity has some intuitive meaning: Since a proportion α of the bits are lost in the channel, we can recover (at most) a proportion $1 - \alpha$ of the bits.
- Hence the capacity is at most $1 - \alpha$. It is not immediately obvious that it is possible to achieve this rate

Example

- The probability distribution of a binary info source is $p(X) = \{p(0) = p, p(1) = 1 - p\}$. The transition probabilities of the channel is $p(Y/X) = \begin{bmatrix} 1 - \epsilon & \epsilon \\ \epsilon & 1 - \epsilon \end{bmatrix}$.
- Please calculate the mutual info $I(X; Y)$.
 - Please calculate the channel capacity and the corresponding optimal probability distribution of the binary info source $p^*(X)$.

Example

- The probability distribution of a binary info source is $p(X) = \{p(0) = p, p(1) = 1 - p\}$. The transition probabilities of the channel is $p(Y/X) = \begin{bmatrix} 1 - \alpha - \epsilon & \alpha & \epsilon \\ \epsilon & \alpha & 1 - \alpha - \epsilon \end{bmatrix}$.
- Please calculate the mutual info $I(X; Y)$.
 - Please calculate the channel capacity and the corresponding optimal probability distribution of the binary info source $p^*(X)$.

Example

- We have a binary info source X having a distribution of $\{p(x = 0) = 0.6, p(x = 1) = 0.4\}$ and a binary symmetric channel having transition probabilities of $p(Y/X) = \begin{bmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{bmatrix}$. If the messages of X 's expanded source of order 2 are transmitted in the corresponding expanded channel of order 2, please calculate the mutual information and the capacity of the expanded channel.

Example

- The probability distribution of a binary info source is $p(X) = \{p(0) = 0.8, p(1) = 0.2\}$. The transition probabilities of the channel is $p(Y/X) = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.1 & 0.3 & 0.6 \end{bmatrix}$.
- Please calculate the mutual info $I(X; Y)$.
 - Please calculate the channel capacity and the corresponding optimal probability distribution of the binary info source $p^*(X)$.

Capacity of A Generic Channel

Step 1: By $\sum_{j=1}^m p(b_j/a_i)\beta_j = \sum_{j=1}^m p(b_j/a_i)\log_2 p(b_j/a_i)$, get β_j ;

Step 2: By $C = \log_2 \sum_{j=1}^m 2^{\beta_j}$, get C ;

Step 3: By $p(b_j) = 2^{\beta_j - C}$, get $p(b_j)$;

Step 4: By $p(b_j) = \sum_{i=1}^n p(a_i)p(b_j/a_i)$, get $p(a_i)$;

Calculation Method for Capacity of A Generic Discrete Channel

- Could you solve the maximization?

$$C = \max_{\{p(a_i)\}} I(X; Y) = I(\{p^*(a_i)\}, \{p(b_j/a_i)\}) \text{ bit/sign}$$

- Lagrangian extremum method

$$\max I(Y; X) \text{ s.t. } \sum_{i=1}^n p(x_i) = 1$$

$$\text{Let } \frac{\partial}{\partial p(x_k)} \{I(Y; X) - \lambda [\sum_{i=1}^n p(x_i) - 1]\} = 0, k = 1, 2, \dots, n$$

$$\text{Then we get } \frac{\partial}{\partial p(x_k)} \left\{ -\sum_{j=1}^m p(y_j) \log p(y_j) \right.$$

$$\left. + \sum_{i=1}^n \sum_{j=1}^m p(x_i) p(y_j / x_i) \log p(y_j / x_i) - \lambda [\sum_{i=1}^n p(x_i) - 1] \right\} = 0$$

$$k = 1, 2, \dots, n$$

$$\log(x) = \log_2(x)$$

Calculation Method for Capacity of A Generic Discrete Channel(2)

$$\therefore \frac{\partial p(y_j)}{\partial p(x_k)} = \frac{\partial}{\partial p(x_k)} \sum_{i=1}^n p(x_i) p(y_j / x_i) = p(y_j / x_k), k = 1, 2, \dots, n$$

Differential of 1st item

$$\begin{aligned} \therefore \frac{\partial}{\partial p(x_k)} \left\{ - \sum_{j=1}^m p(y_j) \text{lb} p(y_j) \right\} \\ = - \sum_{j=1}^m [p(y_j / x_k) \text{lb} p(y_j) + \text{lb} p(y_j / x_k)] \end{aligned}$$

$$k = 1, 2, \dots, n$$

Calculation Method for Capacity of A Generic Discrete Channel(3)

Differential of 2nd and 3rd item

$$\frac{\partial}{\partial p(x_k)} \left\{ \sum_{i=1}^n \sum_{j=1}^m p(x_i) p(y_j / x_i) \text{lb} p(y_j / x_i) - \lambda \left[\sum_{i=1}^n p(x_i) - 1 \right] \right\}$$

$$= \sum_{j=1}^m p(y_j / x_k) \text{lb} p(y_j / x_k) - \lambda \quad (k = 1, 2, \dots, n)$$

$$\sum_{j=1}^m \text{lb} p(y_j / x_k) = \text{lb} e$$

$$\text{Then we have } - \sum_{j=1}^m [p(y_j / x_k) \text{lb} p(y_j) + \text{lb} p(y_j / x_k)]$$

$$+ \sum_{j=1}^m p(y_j / x_k) \text{lb} p(y_j / x_k) - \lambda = 0, k = 1, 2, \dots, n$$

Calculation Method for Capacity of A Generic Discrete Channel(4)

$$-\sum_{j=1}^m [p(y_j / x_k) lbp(y_j) + p(y_j / x_k) lbp(y_j / x_k)]$$

$$= lbe + \lambda$$

$$k = 1, 2, \dots, n$$

$$-\sum_{k=1}^n p(x_k) \sum_{j=1}^m [p(y_j / x_k) lbp(y_j) - p(y_j / x_k) lbp(y_j / x_k)]$$

$$= \sum_{k=1}^n p(x_k) [lbe + \lambda]$$

Calculation Method for Capacity of A Generic Discrete Channel(5)

$$\begin{aligned} & - \sum_{k=1}^n \sum_{j=1}^m p(x_k) p(y_j / x_k) \text{lb} p(y_j) \\ & + \sum_{k=1}^n \sum_{j=1}^m p(x_k) p(y_j / x_k) \text{lb} p(y_j / x_k) \\ & = \text{lb} e + \lambda \end{aligned}$$

$$C = \text{lb} e + \lambda$$

Calculation Method for Capacity of A Generic Discrete Channel(6)

$$-\sum_{j=1}^m [p(y_j / x_k) \text{lb} p(y_j) + p(y_j / x_k) \text{lb} p(y_j / x_k)]$$

$$= \text{lb} e + \lambda$$

$$k = 1, 2, \dots, n$$

$$\sum_{j=1}^m p(y_j / x_k) \text{lb} p(y_j / x_k) = \sum_{j=1}^m p(y_j / x_k) \text{lb} p(y_j) + C$$

$$= \sum_{j=1}^m p(y_j / x_k) [\text{lb} p(y_j) + C] = \sum_{j=1}^m p(y_j / x_k) \beta_j$$

$$k = 1, 2, \dots, n$$

$$\text{lb} p(y_j) = \beta_j - C \quad j = 1, 2, \dots, m$$

Calculation Method for Capacity of A Generic Discrete Channel(7)

$$p(y_j) = 2^{\beta_j - C} \quad j = 1, 2, \dots, m$$

$$\sum_{j=1}^m p(y_j) = \sum_{j=1}^m 2^{\beta_j - C} = 2^{-C} \sum_{j=1}^m 2^{\beta_j} = 1$$

$$C = lb\left(\sum_{j=1}^m 2^{\beta_j}\right)$$

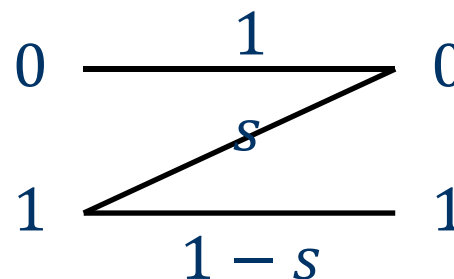
Binary Z-Channel

- Z-Channel: the transition probability of $0 \rightarrow 1$ is zero, while the transition probability of $1 \rightarrow 0$ is s .

Channel Transition Matrix

$$\begin{bmatrix} 1 & 0 \\ s & 1-s \end{bmatrix}$$

Channel Transition Graph



- The capacity of a Z-channel is $C = \log_2 \left[1 + (1-s)s^{\frac{s}{1-s}} \right]$, the corresponding optimal source distribution is

$$p(a_1) = \frac{1 - s^{\frac{1}{1-s}}}{1 + (1-s)^{\frac{s}{1-s}}}$$

$$p(a_2) = \frac{s^{\frac{s}{1-s}}}{1 + (1-s)^{\frac{s}{1-s}}}$$

Example: Z-Channel

- Problem 7.8 in 《Elements of Information》 (Thomas M. Cover and Joy A. Thomas)

7.8 *Z-channel*. The *Z*-channel has binary input and output alphabets and transition probabilities $p(y|x)$ given by the following matrix:

$$Q = \begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \end{bmatrix} \quad x, y \in \{0, 1\}$$

Find the capacity of the *Z*-channel and the maximizing input probability distribution.

Solution:

- First we express $I(X;Y)$, the mutual information between the input and output of the Z-channel, as a function of $x = \Pr(X = 1)$:

$$H(Y|X) = \Pr(X = 0) \cdot 0 + \Pr(X = 1) \cdot 1 = x$$

$$H(Y) = H(\Pr(Y = 1)) = H(x/2)$$

$$I(X;Y) = H(Y) - H(Y|X) = H(x/2) - x$$

- Since $I(X;Y) = 0$ when $x = 0$ and $x = 1$, the maximum mutual information is obtained for some value of x such that $0 < x < 1$.
- Using elementary calculus, we determine that

$$\frac{d}{dx} I(X;Y) = \frac{1}{2} \log_2 \frac{1 - x/2}{x/2} - 1,$$

which is equal to zero for $x = 2/5$. (It is reasonable that $\Pr(X = 1) < 1/2$ because $X = 1$ is the noisy input to the channel.)

- So the capacity of the Z-channel in bits is

$$H(1/5) - 2/5 = 0.722 - 0.4 = 0.322.$$